WSABI Summer Research Lab **Summer 2026**

Lab 4: GLMs and GAMs

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4.1 Batting Averages as a Binomial GLM

4.1.1 Data

We are given player-level batting data from 2020 and 2021. Each row represents one MLB player who had at least 50 at-bats in both seasons. The columns are:

- `playerID`: player identifier;
- `H_2021`: hits in 2021;
- `AB_2021`: at-bats in 2021;
- `BA_2021`: observed batting average in 2021;
- `BA_2020`: observed batting average in 2020.

In Lecture 1, we modeled `BA_2021` directly as a real-valued response. In this lab, you will revisit the same prediction problem using a **binomial GLM** that models hits out of at-bats.

4.1.2 Your Assignment

Task 1: Fit two competing models.

- Fit the day-1 linear regression model

$$\mathbb{E}[BA^{(2021)} \mid BA^{(2020)}] = \beta_0 + \beta_1 BA^{(2020)}.$$

- Fit the binomial GLM

$$H_i^{(2021)} \mid BA_i^{(2020)} \sim \text{Binomial}(AB_i^{(2021)}, p_i), \qquad \text{logit}(p_i) = \alpha_0 + \alpha_1 BA_i^{(2020)}.$$

- Write down both fitted models.

Task 2: Compare the fitted mean curves.

- Plot `BA_2021` against `BA_2020`.
- Overlay the fitted mean from the linear regression model.
- Overlay the fitted mean from the binomial GLM.

- Make the plot reflect how many at-bats each player had in 2021.
- Briefly explain how the two fits differ and why the binomial GLM is a more natural model for hits and outs.

Task 3: Interpret the binomial GLM.

- Report the coefficient estimates, standard errors, and a 95% confidence interval for the GLM coefficients.
- Interpret the slope on the **log-odds** scale.
- Translate the slope into the effect of a 0.010 increase in 2020 batting average.

Task 4: Show why the denominator matters.

- Pick a hypothetical player with `BA_2020 = 0.260`.
- Use your fitted GLM to estimate that player's 2021 hit probability p.
- Compare two workloads:
  (a) 60 at-bats;
  (b) 600 at-bats.
- For each workload, report the expected number of hits.
- For each workload, compute an approximate 95% interval for the **observed batting average** using

$$p \pm 1.96\sqrt{\frac{p(1-p)}{AB}}.$$

- Explain why the low-at-bat player has a much wider interval even though the underlying hit probability is the same.

4.2 Field-Goal Success with a Logistic GAM

4.2.1 Data

We are given field-goal attempts from NFL play-by-play data. Each row represents one kick, and the columns are:

- `fg_made`: 1 if the kick was made and 0 otherwise;
- `ydl`: yard line, measured from the opponent's end zone;
- `kq`: kicker quality;
- `kicker`: kicker name.

This is a probability problem, so all models in this section should be **binomial** models.

4.2.2 Your Assignment

Task 1: Fit competing probability models.

- Fit at least one simple logistic GLM in `ydl`.
- Fit at least one richer logistic GLM, such as a quadratic model and/or a model including `kq`. Write down the exact model form you used.
- Fit a logistic GAM of the form

$$\text{logit}(p_i) = \beta_0 + f(ydl_i) + \gamma kq_i.$$

Task 2: Compare the models out of sample.

- Use a train/test split or cross-validation.
- Compare the models using **log loss** as your main metric.
- State which model you prefer and why.

Task 3: Interpret the GAM fit.

- Report the estimated coefficient for `kq`, together with its standard error and a 95% confidence interval.
- Report the effective degrees of freedom (edf) of the smooth term.
- Explain what it means if the edf is noticeably larger than 1.

Task 4: Visualize the fitted curves.

- Plot predicted make probability against `ydl` for your preferred GLM and your preferred GAM.
- Also compute binned observed make rates and add them to the plot.
- Comment on where the GAM improves on the simpler GLM and where the two fits are similar.

Task 5: Make concrete predictions.

- For a kicker with league-median `kq`, estimate make probability at
  (a) 20 yards from the opponent's end zone;
  (b) 35 yards from the opponent's end zone;
  (c) 50 yards from the opponent's end zone.
- For at least one of these yard lines, compute an approximate 95% confidence interval for the fitted probability by first building the interval on the logit scale and then transforming back with `plogis()`.

Task 6: Reflect.

- Explain the difference between choosing polynomial terms by hand in a GLM and letting a GAM learn a smooth curve.
- Give one reason a GAM can help.
- Give one reason a GAM can hurt.